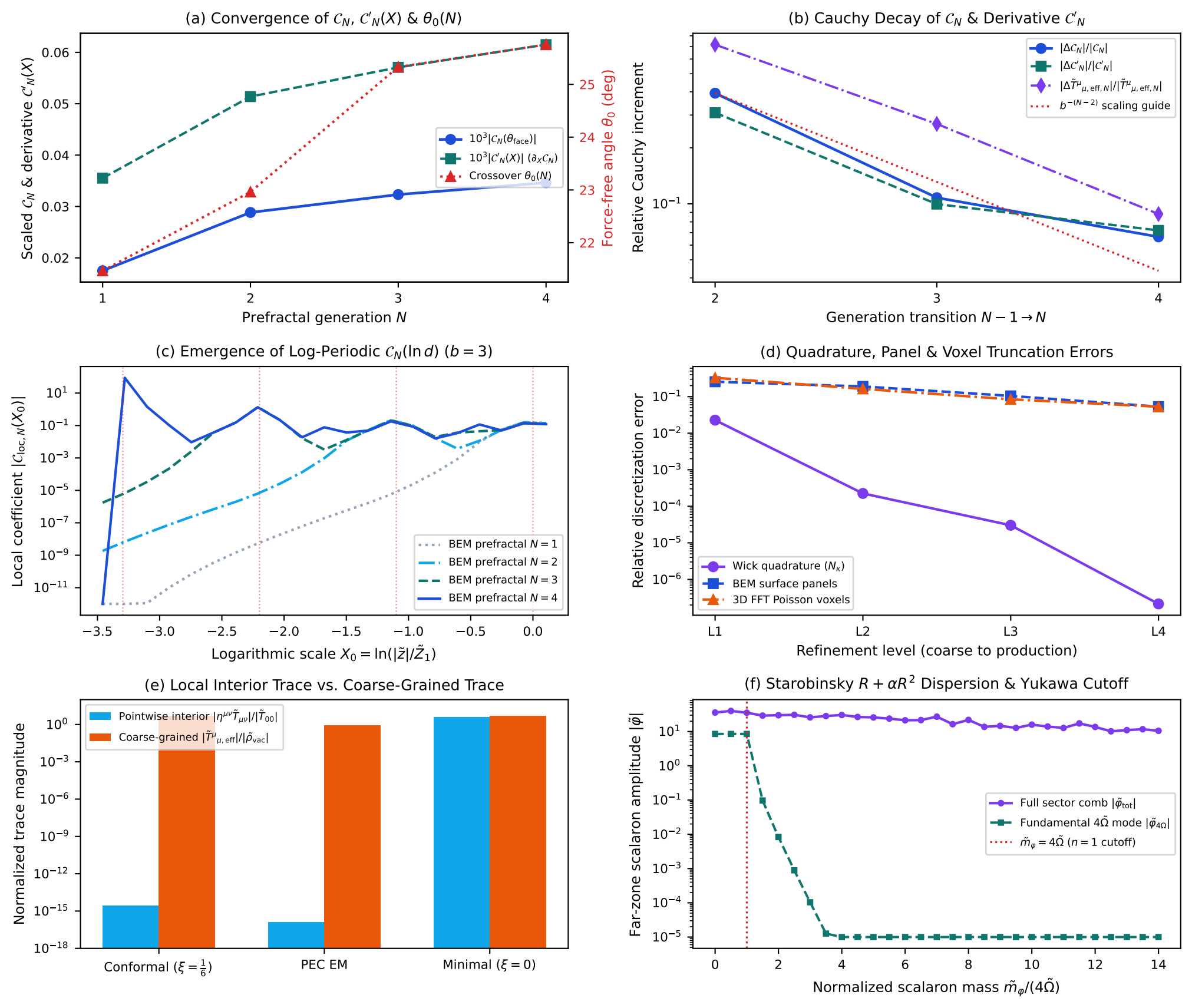
(a) Convergence of $\mathcal{C}_N$, $\mathcal{C}'_N(X)$ & $\theta_0(N)$
Scaled $\mathcal{C}_N$ & derivative $\mathcal{C}'_N(X)$
Force-free angle θ_0 (deg)
Prefractal generation N
$10^3|\mathcal{C}_N(\theta_{\text{face}})|$
$10^3|\mathcal{C}'_N(X)|$ $(\partial_X \mathcal{C}_N)$
Crossover $\theta_0(N)$
(b) Cauchy Decay of $\mathcal{C}_N$ & Derivative $\mathcal{C}'_N$
Relative Cauchy increment
Generation transition $N-1 \to N$
$|\Delta \mathcal{C}_N|/|\mathcal{C}_N|$
$|\Delta \mathcal{C}'_N|/|\mathcal{C}'_N|$
$|\Delta \tilde{T}^{\mu}_{\mu,\text{eff},N}|/|\tilde{T}^{\mu}_{\mu,\text{eff},N}|$
$b^{-(N-2)}$ scaling guide
(c) Emergence of Log-Periodic $\mathcal{C}_N(\ln d)$ $(b=3)$
Local coefficient $|\mathcal{C}_{\text{loc},N}(X_0)|$
Logarithmic scale $X_0 = \ln(|\tilde{z}|/\tilde{Z}_1)$
BEM prefractal $N=1$
BEM prefractal $N=2$
BEM prefractal $N=3$
BEM prefractal $N=4$
(d) Quadrature, Panel & Voxel Truncation Errors
Relative discretization error
Refinement level (coarse to production)
Wick quadrature (N_κ)
BEM surface panels
3D FFT Poisson voxels
(e) Local Interior Trace vs. Coarse-Grained Trace
Normalized trace magnitude
Pointwise interior $|\eta^{\mu\nu}\tilde{T}_{\mu\nu}|/|\tilde{T}_{00}|$
Coarse-grained $|\tilde{T}^{\mu}_{\mu,\text{eff}}|/|\tilde{\rho}_{\text{vac}}|$
Conformal ($\xi=\frac{1}{6}$)
PEC EM
Minimal ($\xi=0$)
(f) Starobinsky $R+\alpha R^2$ Dispersion & Yukawa Cutoff
Far-zone scalaron amplitude $|\tilde{\varphi}|$
Normalized scalaron mass $\tilde{m}_\varphi/(4\tilde{\Omega})$
Full sector comb $|\tilde{\varphi}_{\text{tot}}|$
Fundamental $4\tilde{\Omega}$ mode $|\tilde{\varphi}_{4\Omega}|$
$\tilde{m}_\varphi = 4\tilde{\Omega}$ ($n=1$ cutoff)